

Improving Patient Appointment Attendance and Healthcare Support Using Data & AI

Validated dashboard findings, testing evidence, cross-track alignment and actionable recommendations

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Dataset: 5,000 appointment records

HealthConnect Appointment & Patient Insights

1. Business Problem & Project Objective

What the analysis is designed to understand and improve

48.5%

NO-SHOW RATE

46.3%

ATTENDANCE RATE

47.6%

WITH REMINDER

42.7%

WITHOUT REMINDER

24.2

AVG WAITING TIME

Why this matters

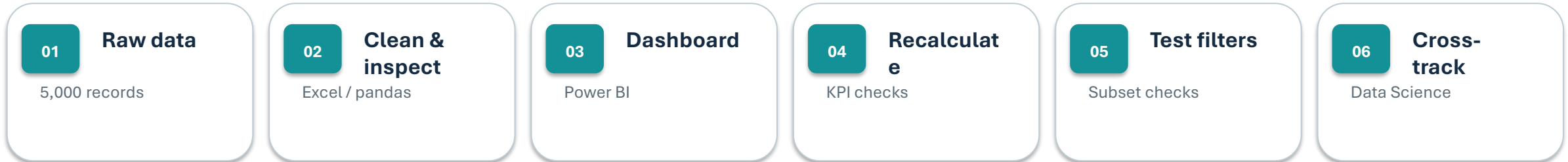
- Missed appointments reduce effective use of available clinical slots.
- A high no-show level can create avoidable scheduling and patient-support pressure.
- The analysis identifies where attendance risk is concentrated rather than treating all patients the same.
- Insights support targeted reminders, access support and operational decisions.

Core questions

- Which factors are most associated with no-shows?
- Does receiving a reminder correspond with higher attendance?
- Which reminder channel performs best overall?
- Which patient segments carry the highest attendance risk?
- Does distance become more important when combined with booking lead time?

2. Data, Tools & Analytical Method

Week 7 testing used the raw CSV to independently validate dashboard outputs



Key analytical fields

Distance to clinic

Booking lead time

Reminder channel

Age group

Appointment type

Waiting time

Week 7 validation standard

- Rate KPI difference ≤ 0.1 percentage points.
- Filters checked against expected direction and manual subsets.
- Channel ranking checked in aggregate and within distance bands.
- Highest-risk combination retested across appointment types.
- Cross-track results compared with Data Science findings.

3. Validated Key Findings

Findings taken directly from the Week 7 validated dashboard analysis

Distance is the strongest single driver

Very Far **68.1% no-show**



Close **46.5% no-show**



21.6 percentage-point spread

The analysis shows a clear attendance-risk gradient as distance increases.

Risk increases further when factors combine

VERY FAR + 2-4 WEEKS

83.3% NO-SHOW

This elevated risk persisted across all four appointment types tested.

Diagnostic Test

Follow-up

General

Specialist

Interpretation: distance and booking lead time should be considered together for targeted risk identification.

4. Reminder & Channel Performance

Reminder exposure is associated with a modest attendance improvement

47.6%

ATTENDANCE WITH REMINDER

42.7%

ATTENDANCE WITHOUT
REMINDER

4.9 pp

REMINDER GAP

Overall channel ranking



Important Week 7 caveat

Moderate Distance **Email** v **SMS**
 47.1% s **46.4%**

Email is marginally higher by 0.7 pp in this distance band.

- SMS remains the overall default based on the validated aggregate ranking.
- The Moderate Distance difference is small enough to justify an A/B test rather than a permanent channel change.

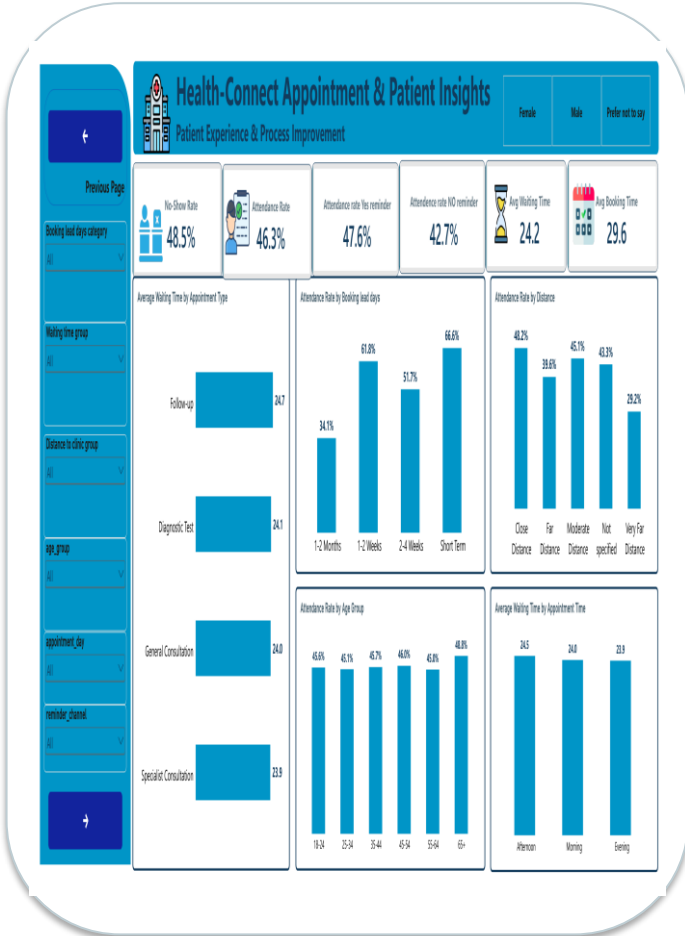
5. Dashboard Walkthrough

Three dashboard pages translate the analysis into an operational view



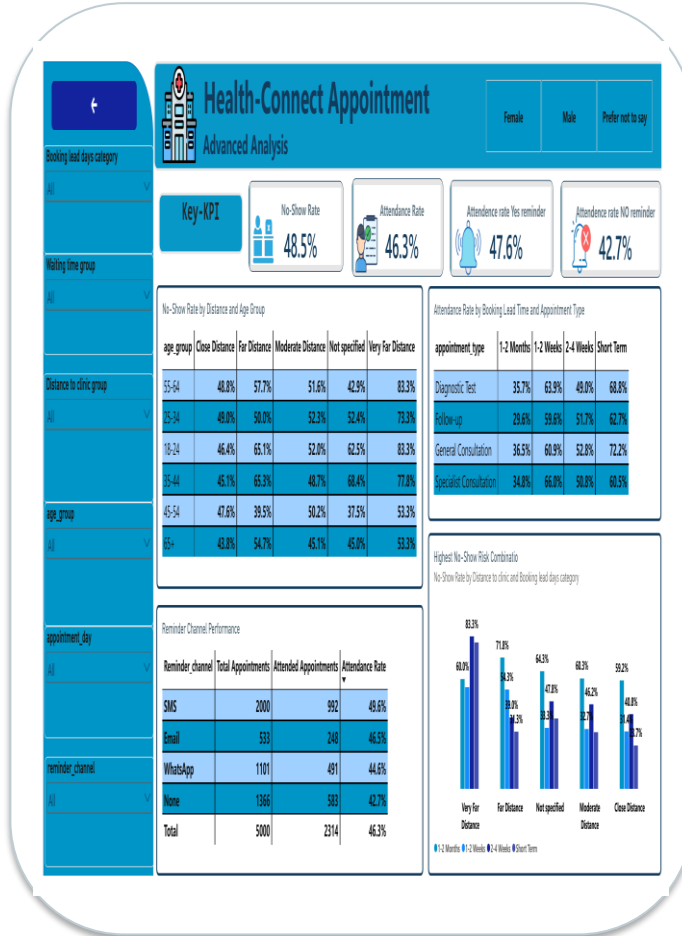
Page 1

Attendance & No-Show Analysis



Page 2

Patient Experience & Process



Page 3

Advanced Analysis

6. Validation & Testing Evidence

Week 7 moved the dashboard from visual reporting to tested analytical evidence

5 / 5 PASS

KPI TESTS

7 / 7 PASS

FILTER TESTS

0.0 pp

KPI VARIANCE

83.3%

RISK COMBINATION

5,000

BASELINE

What was tested

- Attendance, no-show and reminder KPIs were independently recalculated.
- Distance, age, reminder channel, appointment type and booking lead-time filters were tested.
- Filter reset returned baseline KPIs with no filter leakage detected.
- Channel ranking was rechecked across distance bands.
- Very Far + 2–4 Weeks was tested across all four appointment types.

Validated outcomes

- All five KPI tests passed with zero variance.
- All seven filter tests passed.
- SMS ranked first overall for attendance at 49.6%.
- The 83.3% risk combination persisted across appointment types.
- The copied-value error in the original Test Table was corrected and refined visuals were retested against the raw CSV.

7. Cross-Track Collaboration with Data Science

Analytics findings were compared with the Data Science feature analysis

Shared evidence

Baseline no-show	48.5%
Very Far no-show	68.1%
SMS attendance	49.6%
Very Far + 2–4 Weeks	83.3%

Cross-track alignment

- Data Science identified Booking Lead Time and Distance among the top two model features.
- Analytics independently identified the same two strongest drivers.
- The artefact column Unnamed: 0 was removed before model rerun.
- The cleaned Data Science dataset matched the dashboard population of 5,000 records.
- Removing the artefact did not change the top two features.

Week 8 dependency: final Data Science run on the refined dashboard outputs remains to be completed.

8. Recommendations for HealthConnect

Recommendations are tied directly to the validated risk patterns

HIGH

Target Very Far + 2–4 Weeks

Prioritise intensive reminders and/or transport support for the 83.3% no-show segment.

HIGH

Use SMS as default

SMS recorded the highest overall attendance rate at 49.6%.

MEDIUM

A/B test Email in Moderate Distance

Email was 47.1% vs SMS 46.4%; the gap is only 0.7 pp.

MEDIUM

Encourage shorter booking lead times

Where clinically appropriate, shorter lead times showed higher attendance than longer booking windows.

MEDIUM

Offer access support

Consider telehealth or transport support for Very Far patients because distance is the strongest single driver.

LOW

Use age as a supporting signal

Age alone showed a relatively narrow no-show range; use it with distance and lead time.

9. Limitations, Caveats & Week 8 Next Steps

What the analysis can support — and what still requires further testing

Limitations / open issues

- Moderate Distance has a marginal Email-over-SMS difference that needs A/B testing.
- Not specified distance (52.2% no-show) may hide an important access or data-quality factor.
- Cross-page navigation/filter interaction on the mobile layout was not yet tested.
- The final Data Science model run on Week 7 refined dashboard outputs remains a dependency.

Week 8 next steps

- Run an Email vs SMS A/B test for the Moderate Distance segment.
- Investigate and improve capture of missing distance information.
- Complete the final Data Science run using the refined Week 7 outputs.
- Test mobile dashboard navigation and filter interactions.
- Track whether targeted interventions improve attendance over time.

10. Final Takeaway

A validated, targeted approach is more actionable than a one-size-fits-all reminder strategy

48.5% no-show is not driven by one factor alone.

DISTANCE

68.1%

Very Far no-show rate

COMBINATION

83.3%

Very Far + 2–4 Weeks

REMINDERS

+4.9 pp

Attendance gap with reminders

Recommended direction: target the highest-risk access segment, keep SMS as the overall default, test the Moderate Distance exception, and monitor results.

Thank you